Research Directions

We are broadly interested in **Control & Optimization**, **Machine Learning**, and **Sustainable Energy Systems**. Current specific research directions in COLAS Lab include:

- Data-driven modeling, control and learning for dynamical systems
- Control-inspired machine learning with structural guarantees in generative AI
- Power flow analysis, control, and optimization in power networks

My mentoring style

- Each student will have a weekly one-to-one meeting with me for research discussions (typically 0.5 to 1 hour). I may also organize a weekly group-gathering event, where all the group members can have a cup of coffee together, share recent ideas/findings (typically 1 to 1.5 hours). Except for those, I will let students flexibly arrange their schedules, and try not let anything distract them from their focus on courseworks and research.
- I strongly encourage students to attend international conferences (e.g., Control Conferences: CDC, ACC, L4DC; AI/ML Conferences: NeurIPS, ICML, ICLR; Power Systems Conferences: ACM e-Energy, PSCC, PESGM), and present their works. I will try my best to find funding resources to cover the cost of attendance. I also strongly encourage students to take short-term visits to other research groups at top oversea universities and pursue industry internships in high-tech companies.
- For junior students who newly join the group, I will typically propose a research problem with relatively mature idea, according to the students' primary background. Once they have their first paper published, I will let them explore research problems more freely, so long as their exploration stays within the broaden research interests of our group (Control & Optimization; AI/ML; Power Systems).

How to reach out to me

For PhD applicants: Send me an email with your CV, transcripts, and anything you think might help, with the email subject as [PhD Application]–[Your Name]. The monthly stipend is 15K RMB. See here for the instructions from HKUST Guangzhou.

For MPhil applicants: You need to first apply to the Red Bird MPhil Program. Once admitted, send me an email with your CV and the email subject as [MPhil Application]–[Your Name]. The monthly stipend is 10K RMB (provided by the university).

For RA/Visiting Student: Send me an email with your CV and indicate how long you will stay with us, with the email subject as [RA/Visiting Application]–[Your Name]. The monthly stipend will be on a case-by-case basis. See here for the instructions from HKUST Guangzhou.

Note: My current email address: `zhenyiyuan@hkust-gz.edu.cn`. I may not be able to respond to all the application emails, but I promise that I will carefully review all the applications.